HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Batch: 29th | Semester: Fall 2025

Final Examination

Date: 03/03/2026 | Time: 2 hours | Total Marks: 40

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5 x 8 = 40

Answer any **five** questions.

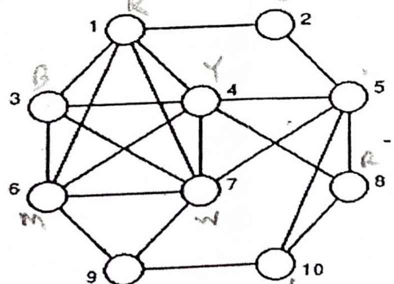
1. (a) Write the applications of trees. — CLO 4 — 2
   (b) Define lattices, Infimum lattices and Supremum lattices. — CLO 4 — 3
   (c) Prove that the set of even integers forms Abelian group. — CLO 4 — 3

2. (a) State the steps of constructing Hasse Diagram from the Partial Ordering. — CLO 4 — 2
   (b) Let A = {2, 3, 6, 24, 48, 96, 192} be a set. Find out R = {(a, b) | a divides b}. Draw the partial ordering diagram for R and finalize the Hasse diagram from the partial ordering diagram, also find out the maximal and minimal. — CLO 4 — 6

3. (a) State the differences between Directed and Undirected graph in terms of representation in adjacency matrix. — CLO 4 — 3
   (b) Explain the representation of a graph in memory with advantages and disadvantages. — CLO 4 — 5

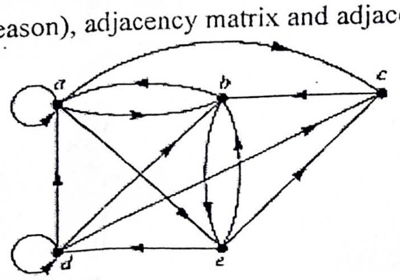
4. (a) State the corollaries for planar graph. — CLO 4 — 3
   (b) Define Composite relation. Let A = {1, 2, 3}, B = {1, 2, 3, 4} and C = {0, 1, 2} be 3 sets. R be a relation from A to B and S be a relation from B to C, where R = {(a, b) | a $\leq$ b} and S = {(b, c) | b $\leq$c}. Find R, S, S $\circ$ R, R $\circ$ S. — CLO 4 — 5

5. (a) Color the graph by using graph coloring technique and find the chromatic number. — CLO 4 — 3



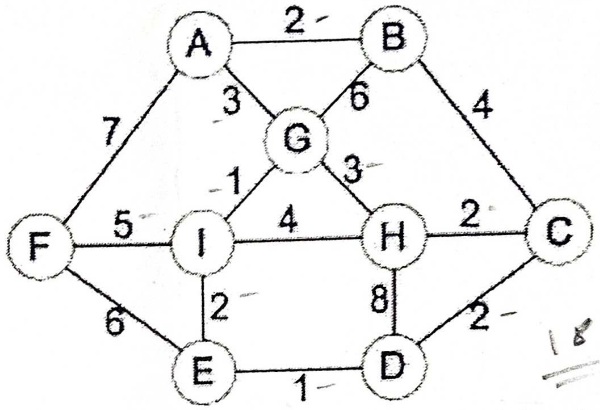
   (b) Let $A = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 5 & 4 \\ 2 & 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 5 & 4 \\ 2 & 3 & 2 \end{bmatrix}$ be 3×3 matrices. Find A + B, A x B and the inverse of the matrix B. — CLO 4 — 5

6. (a) Define fallacy and congruence relation. — CLO 1 — 2
   CLO 4
   (b) Find the properties (with reason), adjacency matrix and adjacency list for the following di-graph: — CLO 4 — 6

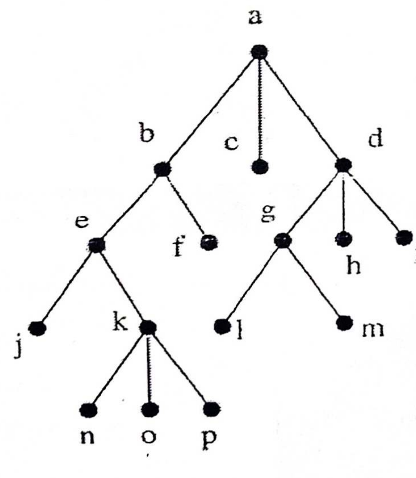


Determine degree, in-degree and out-degree of each vertex for the above graph.

7. (a) Find the minimum spanning tree for the following weighted graph by using CLO 4 4
(i) Prim's algorithm
(ii) Kruskal's algorithm



(b) Briefly describe tree traversal techniques and find the order of the elements using tree traversal techniques after traversing the following tree: CLO 4 4



HAMDARD UNIVERSITY BANGLADESH

Department of Electrical and Electronics Engineering
Faculty of Science, Engineering and Technology

Course Code: EEE 117 | **Course Title: Electrical Engineering**
Final Examination | Semester: Fall 2025(Day)
Date: 5/3/2026 | Total Time: 2 Hours | **Total Marks: 40**

Answer any three (04) out of the following questions: $4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

1. (a) Define the following terms: 2 CLO1
   I. Capacitor
   II. Inductor

   (b) Derive the formula for the equivalent capacitance of n capacitors in parallel using proper figure; repeat your answer when inductors are connected in parallel. 4 CLO2

   (c) For the circuit in Figure-1(c), find the voltage across each capacitor. 4 CLO3

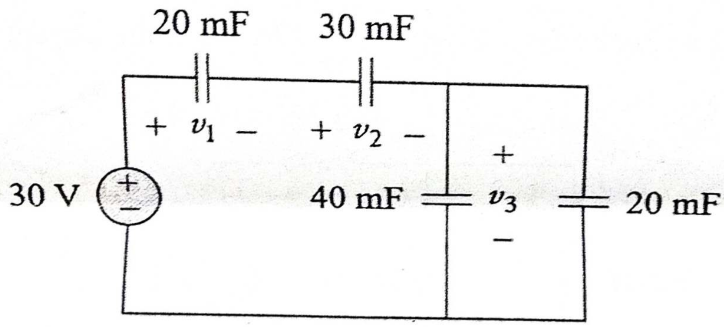


Figure-1(c)

2. (a) Note down the formulas of current, voltage and power across the Capacitor and Inductor separately. 2 CLO1

   (b) An initially uncharged 1-mF capacitor has the current shown in Figure 2(b) across it. Calculate the voltage across it at t = 2 ms and t = 5 ms. 4 CLO3

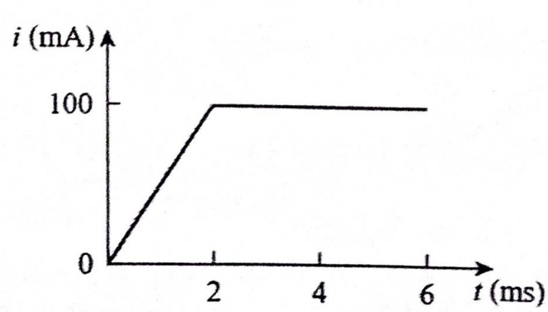


Figure-2(b)

(c) For the circuit in Figure-2(c), $i(t)=4(2-e^{-10t})$ mA. If $i_2(0) = -1$ mA find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$. 4 CLO3

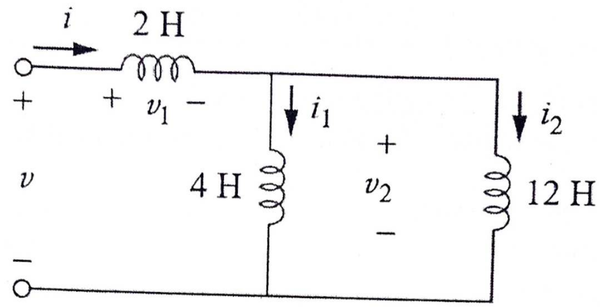


Figure-2(c)

3. (a) Write a short note on the different types of both RL & RC Circuit. 2 CLO1

(b) Derive the source free response of a RC Circuit and also sketch the current decay curve and explain how the inductor current behaves over time. 4 CLO2

(c) For the circuit in Figure-3(c), Given $i(0) = 10A$; Find the current $i(t)$ and $i_x(t)$ 4 CLO3

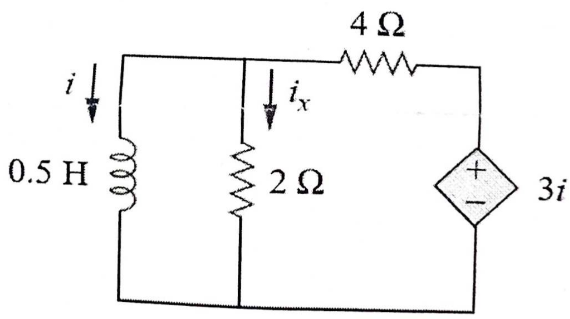


Figure-3(c)

4. (a) Define the followings: 2 CLO1
   (i) Sinusoid
   (ii) Phasor

(b) Evaluate the following complex Numbers: 3 CLO3

(i) $(5 + j2)(-1 + j4) - 5\angle 60°]^*$

(ii) $$\frac{10 + j5 + 3\angle 40°}{-3 + j4} + 10\angle 30° + j5$$

(c) Determine v(t) and i(t) for the circuit has been given below: 5 CLO3

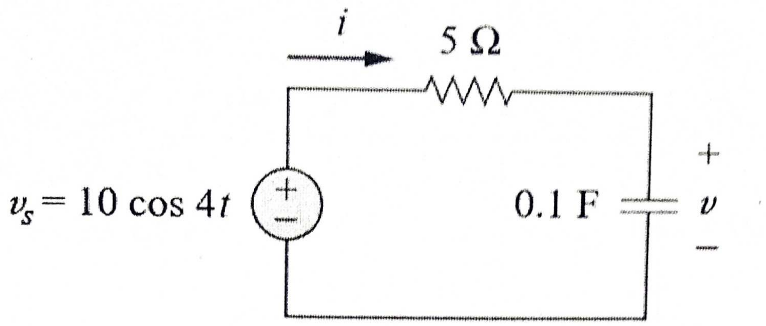


Figure-4(c)

5. (a) Write down the phasor relationships for various circuit elements and draw the proper phasor diagrams. 2 CLO1

(b) Determine $V_0(t)$ for the given circuit. 4 CLO3

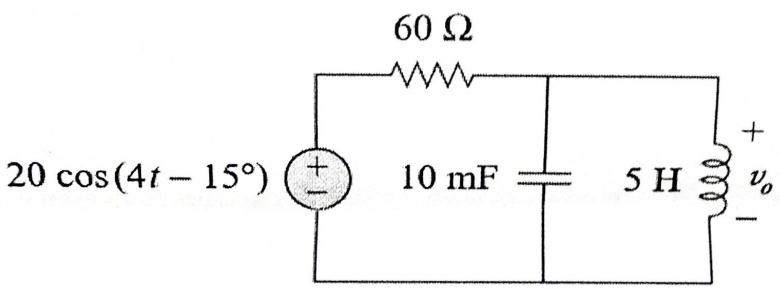


Figure-5(b)

(c) Find current **I** in the circuit of Figure-5(c). 4 CLO3

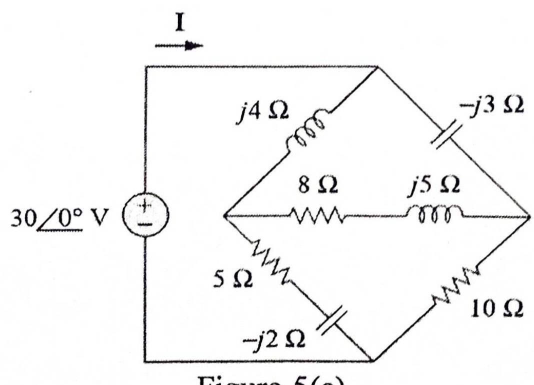


Figure-5(c)

6. (a) Derive the relation between magnetic flux, magnetomotive force (mmf) and reluctance in a magnetic circuit. 2 CLO2

(b) Using nodal analysis, find v_1 and v_2 in the circuit of Figure-6(b). 4 CLO

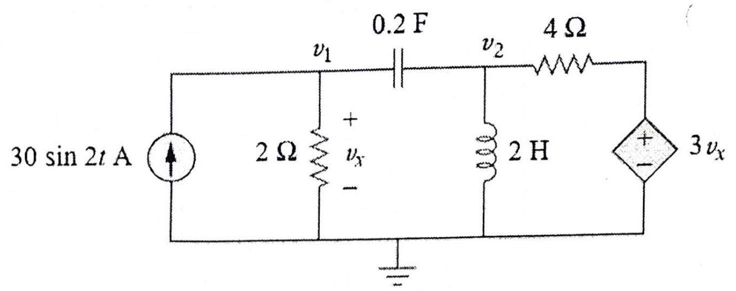


Figure-6(b)

(c) Apply the analogy between electric and magnetic circuits to complete a table with related terms and symbols. 4 CLO3

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY 111 | **Course Title: Physics I**
Final Examination | Semester: Fall 2025
Date: 8/3/2026 | Total Time: 2.00 Hours | **Total Marks: 40**

Answer any four (04) out of the following questions: **$4 \times 10 = 40$**
[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

			CLOs	Marks
Q1.	**(a)**	From the interference due to transmitted light on thin films, find the equation of path difference for fringes.	CLO2	**[6]**
	(b)	Newton's rings are observed in reflected light of $\lambda = 5.9 \times 10^{-5}$ cm. The diameter of the 10^{th} dark ring is 0.5 cm. Find the radius of the curvature of the lens and the thickness of the film.	CLO3	**[4]**
Q2.	**(a)**	From the Fraunhofer diffraction at a circular aperture, show that $x = \frac{f\lambda}{d}$ and also sketch the intensity distribution on the screen.	CLO2	**[6]** 7
	(b)	In a Fraunhofer diffraction due to a narrow slit, a screen is placed 2 m away from the lens to obtain the pattern. If the slit width is 0.2 mm and the first minima lies 5 mm on either side of the central maximum, find the wavelength of light.	CLO3	**[4]**
Q3.	**(a)**	What do you mean by mean free path? – explain.	CLO1	**[2]**
	(b)	From the Brewster's Law of polarization, show that the reflected and the refracted rays are at right angles to each other. Also find the polarization angle for refractive index of 1.52 from the Brewster's Law.	CLO2	**[4]**
	(c)	Newton's rings are formed by reflected light of wavelength 5895 Å with a liquid between the plane and curved surfaces. If the diameter of the 5^{th} ring is 3mm and the radius of curvature of the curved surface is 100 cm, calculate the reflective index of the liquid.	CLO3	**[4]**
Q4.	**(a)**	Write down the postulates of the kinetic theory of gases.	CLO1	**[3]**
	(b)	Show that the energy associated with each degree of freedom is $\frac{1}{2}kT$.	CLO2	**[4]**
	(c)	With what speed would one gram molecule of oxygen at 300k be moving in order that the translational kinetic energy of its centre of mass is equal to the total random kinetic energy of all its molecules?	CLO3	**[3]**

Q5. **(a)** Show that the R.M.S velocity of a gas $C = \sqrt{\frac{3P}{\rho}}$. CLO2 [6]

(b) When a system is taken from the state A to the state B, along the path ACB, 80 J of heat flows into the system, and the system does 30 J of work given in the figure 5.1. CLO3 [4

(i) How much heat flows into the system along the path ADB, if the work done is 10 J.

(ii) The system is returned from the state B to the state A along the curved path. The work done on the system is 20 J. Does the system absorb or liberate heat and how much?

(iii) If $U_A = 0$, $U_D = 40$ J, find the heat absorbed in the process AD and DB.

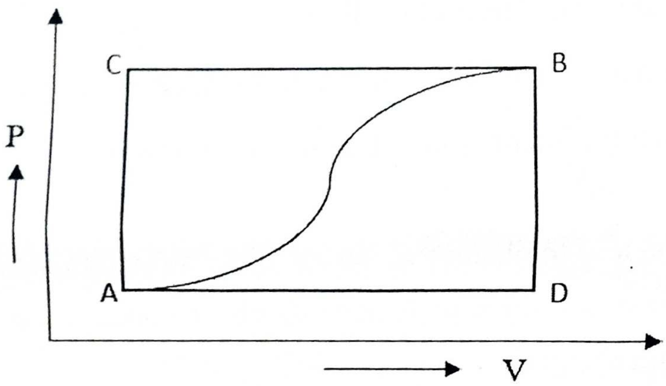


Fig: 5.1

Q6. **(a)** State the first law of thermodynamics. CLO1 **[2]** 2

(b) Show that the amount of heat energy supplied to a system in an isothermal reversible process and is equal to the sum of the work done by the system and the increase in its internal energy. CLO2 **[4]**

(c) The number of molecules per cc of a gas is 2.7×10^{19} at N.T.P. Calculate the number of molecules per cc of the gas. CLO3 **[4]**

(i) at 0° C and 10^{-6} mm pressure of mercury and

(ii) at 39° C and 10^{-6} mm pressure of mercury

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

ENG 111: English Basics

Final Examination
Fall 2025

29th Batch | 1st Semester

10 March 2026 | **Time: 02 Hours**

Total Marks: 40

CLOs	**Question/Statement**	**Marks**
CLO5	1. Rewrite following paragraph using **punctuation and capitalization**:	**08**

Nowadays cyber threats are increasing rapidly, because hackers use advanced tools for social engineering therefore,everyone must stay careful online. Experts believe that using strong passwords is vital but they emphasize one major weakness human error employees often forget their passwords and share sensitive data with strangers mr david tattersall a cyber-security specialist warns that neglecting digital hygiene puts companies sensitive information at risk.

CLO2	2. **Summarize** the following passage:	**10**

Nutrient deficiency is one of the biggest health challenges facing people today, affecting how both the body and the brain develop. It happens mainly because people don't get enough essential things like iron, protein, omega-3 fatty acids, and healthy fats, which the body needs to work properly. These nutrients act as the basic building blocks for our cells and provide the fuel for our metabolism. Usually, these deficiencies come from eating the same few foods every day, relying too much on processed junk food, or not being able to afford fresh ingredients like meat, fish, and vegetables. When the body doesn't get these nutrients, the results are serious, leading to constant tiredness, a weak immune system, and slow growth in children. Specifically, a lack of iron is a major problem because it leads to anemia, a condition where your blood can't carry enough oxygen. This makes a person feel exhausted and worn out all the time, even if they have slept well. Protein is also vital because it helps repair muscles and tissues; without it, the body starts to break down its own muscle for energy. Furthermore, Omega-3s and healthy fats are necessary for the brain to send signals correctly. Since the brain is mostly made of fat, not eating enough healthy fats can lead to brain fog and make it hard to think clearly. Essentially, if you don't eat enough good fats and lipids, your brain and hormones won't function the way they should. This means a person can eat a lot of calories and feel full, but their body is still starving for the actual nutrients it needs to stay healthy.

CLO4	3. Write a formal **cover letter** requesting for the position of Graphics Designer at *Meghna Group of Industries*.	**12**
CLO4	4. Write a paragraph describing **the primary causes of increasing depression among young people.**	**10**

HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT115 | **Course Title: Differential and Integral Calculus**
Final Examination | Semester: Fall 2025
Date: 3/12/2026 | Total Time: 2:00 Hours | Total Marks: 40

Answer any four (04) out of the following questions: $4 \times 10 = 40$
[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No		Question	CLOS	Marks
Q1.	(a)	Define antiderivative with a suitable example.	CLO1	[2]
	(b)	Integrate the following by using method of substitution. (i) $\int [f(x)]^n f'(x)dx$ (ii) $\int \frac{dx}{xl\,ogx}$ (iii) $\int \frac{\sin 2x}{asin^2x+bcos^2x}\,dx$	CLO3	[8]
Q2.	(a)	Briefly explain the definite and indefinite integrals.	CLO1	[3]
	(b)	Evaluate the indefinite integral $\int (1+5x)^3 dx$ by using the method of substitution.	CLO3	[4]
	(c)	Solve the integral $\int \frac{dx}{x\log x}$.	CLO3	[3]
Q3.	(a)	What do you mean by integral curves?	CLO1	[2]
	(b)	Evaluate the integral $\int \frac{1+\cos x}{\sqrt[3]{x+\sin x}}dx$.	CLO3	[4]
	(c)	Solve the integral $\int \frac{\tan x}{(a^2+b^2tan^2x)^2}dx$.	CLO3	[4]
Q4.	(a)	State the three necessary conditions for Rolle's Theorem to be applicable to a function $f(x)$ on a closed interval $[a,b]$.	.CLO2	[3]
	(b)	Verify Rolle's Theorem for the polynomial function $f(x) = x^2 - 5x + 6$ on the interval $[1,4]$.	CLO3	[7]
Q5.	(a)	Evaluate by using product rules: (i) $\int x^2(\ln x)^2dx$ (ii) $\int e^x(\sin x+\cos x)dx$	CLO3	[6]
	(b)	Evaluate $\int \frac{x\,dx}{(x-1)^2(x+2)}$ by using partial fraction.	CLO3	[4]
Q6.	(a)	Show that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.	CLO3	[2]
	(b)	Evaluate $\int_{\frac{\pi}{2}}^{0} sin^5xcos^6x\,dx$.	CLO3	[4]
	(c)	Using beta function evaluate $\int_0^1 x^2(1-x^3)^{\frac{3}{2}}\,dx$.	CLO3	[4]

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 117: Engineering Ethics and Cyber Law

Batch: 29^{th} | Semester: Fall 2025

Time: 2 Hours

Date: 14/03/2026 | **Final Examination** | Total Marks: 40

Answer All questions. $5 \times 8 = 40$

1. (a) What is ransomware? How does it differ from phishing attacks? CLO5 2
   (b) Define cybercrime and explain two main types with examples. CLO5 3
   (c) Demonstrate the key challenges in establishing jurisdiction in cyberspace. CLO4 3

2. A company experiences a data breach because an employee disabled the firewall to speed up access to certain websites. As a result, customer data was leaked online
   (a) Suggest two policies the company could implement to prevent such incidents in the future. CLO5 2
   (b) What legal responsibilities does the company have regarding customer data protection? CLO5 3
   (c) What role does a firewall play in network security, and how could it have prevented this breach? CLO5 3

3. (a) A company has experienced a security breach where an employee used their authorized access to steal confidential data. Analyze how this scenario would be treated under Canadian law, discussing relevant sections of the Criminal Code. CLO2 4

   (b) Describe two key differences between UTM (Unified Threat Management) firewalls and Next-Generation firewalls. Explain which would be more appropriate for a large financial institution and why. CLO2 4

4. A popular social media influencer in Bangladesh has approached your software company with the following requests:
   -To create multiple fake accounts using real people's photos and information.
   -To develop a bot that automatically downloads and reposts viral content from other creators.
   -To implement a system that artificially inflates view counts and engagement metrics.
   (a) Identify which specific provisions of the Digital Security Act 2018 and ICT Act 2006 would be violated by these requests CLO4 4
   (b) As the lead engineer assigned to this project: CLO2 4
   -What ethical considerations should you weigh?
   -How would you handle this situation while maintaining professional

integrity?

5	(a)	What is cyber law, and why is it important in today's digital world?	CLO3	2
	(b)	Discuss two major ethical issues related to software piracy and digital content sharing.	CLO3	3
	(c)	Explain the role of international cyber laws in preventing digital crimes.	CLO4	3
6	(a)	Explain three phases of Hacking	CLO5	4
	(b)	Define five most common types of Cyber Crimes.	CLO5	4
7	(a)	Define Intellectual Property (IP). What are its key types?	CLO3	2
	(b)	Explain the differences between copyright, patent, trademark, and trade secret	CLO3	3
	(c)	How does the Bangladesh Copyright Act, 2000 protect the rights of content creators?	CLO4	3